

Improving Code Quality

Formative Activity

Use some Python code which you have developed in the past and apply at least 3 of the strategies presented at the source to improve its quality.

Code

This formative activity used the Transition Matrix (trans_matsv2.py) code referred to in Unit 8. The code can be found in Appendix 1.

MyPy

MyPy was installed in a virtual environment called 'Uni'. MyPy checks the code for errors. The output can be seen below:

Figure 1: MyPy Testing

```
(uni) PS D:\OneDrive - Smart MAT\2. Uni\Project Management> mypy trans_matsv2.py
Success: no issues found in 1 source file
```

Pylint

While the code itself runs and shows no issues with MyPy there are other tools that can be used to improve code quality. Pylint is a linter that checks coding standards against the PEP8 conventions. This is important as Python code should be highly readable.

Pylint highlighted significant issues with the code found in Appendix 1

Figure 2: Pylint Rating 0.00/10

```
-----
Your code has been rated at 0.00/10
```

There were many instances of errors with formatting concerning trailing white spaces and incorrect indentation. While the indentation did not affect the functionality of the code, it was not compliant with the standard formatting for Python.

Additionally, there were too many local variable in the 'get_expected' function which made the function difficult to read, if it was to be shared with other developers. As a result 2 new functions were created to simplify the code.

This resulted in a 10 out of 10 score.

Figure 3: Pylint 10/10

```
(uni) PS D:\OneDrive - Smart MAT\2. Uni\Project Management> pylint trans_matsv2.py  
-----  
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```

The amended code can be found in Appendix 3 – Pylint Corrections

Flake8

The Flake8 linter was also run on the, now improved, code. Unfortunately, this raises a large number of error which can be seen in appendix 4.

The Flake8 linter highlighted issues which were not picked up by Pylint. Typically, these were spaces around operators and indentations for multi-line statements. Additionally, the length of line limit is reduced when compared to Pylint. There were also expectations regarding block statements and spacing at the end of functions.

Figure 4: Flake8 Corrections

```
● (uni) PS D:\OneDrive - Smart MAT\2. Uni\Project Management> mypy trans_matsv2.py  
Success: no issues found in 1 source file  
● (uni) PS D:\OneDrive - Smart MAT\2. Uni\Project Management> flake8 trans_matsv2.py  
trans_matsv2.py:124:38: E124 closing bracket does not match visual indentation  
trans_matsv2.py:125:37: E124 closing bracket does not match visual indentation  
● (uni) PS D:\OneDrive - Smart MAT\2. Uni\Project Management> flake8 trans_matsv2.py  
● (uni) PS D:\OneDrive - Smart MAT\2. Uni\Project Management> pylint trans_matsv2.py  
-----  
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)  
● (uni) PS D:\OneDrive - Smart MAT\2. Uni\Project Management> flake8 trans_matsv2.py  
○ (uni) PS D:\OneDrive - Smart MAT\2. Uni\Project Management> []
```

Summary

This shows the need to use a range of linters to improve code, both in terms of functionality and readability. One linter is not sufficient as each has separate rules. However, when the amendments were complete, the testing was satisfied for all 3 linters showing that they complement each other and can be used as a coherent suite for improving code.

References

Anaya, M. (2021) *Clean Code in Python: Develop maintainable and efficient code*. 2nd edn. Birmingham: Packt Publishing.


```

#calculates probabilities as a decimal from the sting stored
in TM

chance9 = float(tm['perc_9'][i])/100
chance98 = chance9 + float(tm['perc_8'][i])/100
chance97 = chance98 + float(tm['perc_7'][i])/100
chance96 = chance97 + float(tm['perc_6'][i])/100
chance95 = chance96 + float(tm['perc_5'][i])/100
chance94 = chance95 + float(tm['perc_4'][i])/100
chance93 = chance94 + float(tm['perc_3'][i])/100
chance92 = chance93 + float(tm['perc_2'][i])/100
chance91 = chance92 + float(tm['perc_1'][i])/100
else:
    #caclulates the output for combined-science which has a
different data structure
    expected = round((float(tm['perc_11'][i])*1
        + float(tm['perc_21'][i])*1.5
        + float(tm['perc_22'][i])*2
        + float(tm['perc_32'][i])*2.5
        + float(tm['perc_33'][i])*3
        + float(tm['perc_43'][i])*3.5
        + float(tm['perc_44'][i])*4
        + float(tm['perc_54'][i])*4.5
        + float(tm['perc_55'][i])*5
        + float(tm['perc_65'][i])*5.5
        + float(tm['perc_66'][i])*6
        + float(tm['perc_76'][i])*6.5
        + float(tm['perc_77'][i])*7
        + float(tm['perc_87'][i])*7.5
        + float(tm['perc_88'][i])*8
        + float(tm['perc_98'][i])*8.5
        +
float(tm['perc_99'][i])*9
        )/100,2)

    #caluates cumulative probabilities of achieving a grade
    chance9 = round(float(tm['perc_99'][i])/100,3)
    chance98 = round((chance9 + (float(tm['perc_98'][i]) +
float(tm['perc_88'][i]) )/100),3)
    chance97 = round((chance98 + (float(tm['perc_87'][i]) +
float(tm['perc_77'][i]) )/100),3)
    chance96 = round((chance97 + (float(tm['perc_76'][i]) +
float(tm['perc_66'][i]) )/100),3)
    chance95 = round((chance96 + (float(tm['perc_65'][i]) +
float(tm['perc_55'][i]) )/100),3)
    chance94 = round((chance95 + (float(tm['perc_54'][i]) +
float(tm['perc_44'][i]) )/100),3)
    chance93 = round((chance94 + (float(tm['perc_43'][i]) +
float(tm['perc_33'][i]) )/100),3)

```

```

        chance92 = round((chance93 + (float(tm['perc_32'][i]) +
float(tm['perc_22'][i]) )/100),3)
        chance91 = round((chance92 + (float(tm['perc_21'][i]) +
float(tm['perc_11'][i]) )/100),3)

    #displays probabilities of achieving threshold grades
    print('Chance 7+ ' + str(chance97*100) + '%')
    print('Chance 5+ ' + str(chance95*100) + '%')
    print('Chance 4+ ' + str(chance94*100) + '%')
    #returns statistics as a tuple
    return [expected, round(chance91,3), round(chance92,3),
round(chance93,3), round(chance94,3),
            round(chance95,3), round(chance96,3), round(chance97,3),
round(chance98,3), round(chance9,3), band]

#Only executed if the script is run as a main programme
if __name__ == "__main__":
    #runs the code with test data
    subject='Art and Design'
    #subject = 'Combined Science'
    ks2=102
    get_expected(subject,ks2)

```

Appendix 2 – Pylint on Transition Matrix Code

```
(uni) PS D:\OneDrive - Smart MAT\2. Uni\Project Management> pylint trans_matsv2.py
```

```
***** Module trans_matsv2
```

```
trans_matsv2.py:2:64: C0303: Trailing whitespace (trailing-whitespace)
```

```
trans_matsv2.py:3:0: C0301: Line too long (136/100) (line-too-long)
```

```
trans_matsv2.py:12:0: C0301: Line too long (120/100) (line-too-long)
```

```
trans_matsv2.py:17:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)
```

```
trans_matsv2.py:18:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)
```

```
trans_matsv2.py:21:30: C0303: Trailing whitespace (trailing-whitespace)
```

```
trans_matsv2.py:22:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)
```

```
trans_matsv2.py:24:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)
```

```
trans_matsv2.py:25:44: C0303: Trailing whitespace (trailing-whitespace)
```

```
trans_matsv2.py:25:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)
```

```
trans_matsv2.py:26:0: C0303: Trailing whitespace (trailing-whitespace)
```

```
trans_matsv2.py:29:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)
```

```
trans_matsv2.py:30:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)
```

```
trans_matsv2.py:32:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)
```

```
trans_matsv2.py:34:139: C0303: Trailing whitespace (trailing-whitespace)
```

```
trans_matsv2.py:34:0: C0301: Line too long (139/100) (line-too-long)
```

```
trans_matsv2.py:34:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)
```

trans_matsv2.py:35:0: C0301: Line too long (158/100) (line-too-long)

trans_matsv2.py:37:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:39:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:40:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:41:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:42:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:43:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:44:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:45:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:46:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:47:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:48:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)

trans_matsv2.py:50:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:66:59: C0303: Trailing whitespace (trailing-whitespace)

trans_matsv2.py:68:0: C0303: Trailing whitespace (trailing-whitespace)

trans_matsv2.py:70:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:71:0: C0301: Line too long (104/100) (line-too-long)

trans_matsv2.py:71:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:72:0: C0301: Line too long (105/100) (line-too-long)

trans_matsv2.py:72:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:73:0: C0301: Line too long (105/100) (line-too-long)

trans_matsv2.py:73:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:74:0: C0301: Line too long (105/100) (line-too-long)

trans_matsv2.py:74:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:75:0: C0301: Line too long (105/100) (line-too-long)

trans_matsv2.py:75:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:76:0: C0301: Line too long (105/100) (line-too-long)

trans_matsv2.py:76:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:77:0: C0301: Line too long (105/100) (line-too-long)

trans_matsv2.py:77:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:78:0: C0301: Line too long (105/100) (line-too-long)

trans_matsv2.py:78:0: W0311: Bad indentation. Found 16 spaces, expected 20 (bad-indentation)

trans_matsv2.py:79:0: C0303: Trailing whitespace (trailing-whitespace)

trans_matsv2.py:82:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)

trans_matsv2.py:83:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)

trans_matsv2.py:84:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)

trans_matsv2.py:86:0: C0301: Line too long (107/100) (line-too-long)

trans_matsv2.py:86:0: W0311: Bad indentation. Found 14 spaces, expected 16 (bad-indentation)

trans_matsv2.py:87:0: C0301: Line too long (120/100) (line-too-long)

trans_matsv2.py:89:55: C0303: Trailing whitespace (trailing-whitespace)

trans_matsv2.py:1:0: C0114: Missing module docstring (missing-module-docstring)

trans_matsv2.py:7:0: R0914: Too many local variables (18/15) (too-many-locals)

trans_matsv2.py:7:17: W0621: Redefining name 'subject' from outer scope (line 92) (redefined-outer-name)

trans_matsv2.py:7:26: W0621: Redefining name 'ks2' from outer scope (line 94) (redefined-outer-name)

trans_matsv2.py:7:0: R1710: Either all return statements in a function should return an expression, or none of them should. (inconsistent-return-statements)

trans_matsv2.py:92:4: C0103: Constant name "subject" doesn't conform to UPPER_CASE naming style (invalid-name)

trans_matsv2.py:94:4: C0103: Constant name "ks2" doesn't conform to UPPER_CASE naming style (invalid-name)

Your code has been rated at 0.00/10

Appendix 3 – Pylint Corrections

```
'''
This module caclulates the expected values
for a student studying a given qualification
considering their starting points
'''

import pandas as pd

def chance_gcse(i):
    '''caclulates chances for a GCSE'''
    chance9 = float(tm['perc_9'][i])/100
    chance98 = chance9 + float(tm['perc_8'][i])/100
    chance97 = chance98 + float(tm['perc_7'][i])/100
    chance96 = chance97 + float(tm['perc_6'][i])/100
    chance95 = chance96 + float(tm['perc_5'][i])/100
    chance94 = chance95 + float(tm['perc_4'][i])/100
    chance93 = chance94 + float(tm['perc_3'][i])/100
    chance92 = chance93 + float(tm['perc_2'][i])/100
    chance91 = chance92 + float(tm['perc_1'][i])/100
    chances = [chance91, chance92, chance93,
               chance94, chance95, chance96,
               chance97, chance98, chance9]
    return chances

def chance_combined_science (i):
    '''calculates chances for combined science'''
    chance9 = round(float(tm['perc_99'][i])/100,3)
    chance98 = round((chance9 + (float(tm['perc_98'][i]) +
                                   float(tm['perc_88'][i]) )/100),3)
    chance97 = round((chance98 + (float(tm['perc_87'][i]) +
                                   float(tm['perc_77'][i]) )/100),3)
    chance96 = round((chance97 + (float(tm['perc_76'][i]) +
                                   float(tm['perc_66'][i]) )/100),3)
    chance95 = round((chance96 + (float(tm['perc_65'][i]) +
                                   float(tm['perc_55'][i]) )/100),3)
    chance94 = round((chance95 + (float(tm['perc_54'][i]) +
                                   float(tm['perc_44'][i]) )/100),3)
    chance93 = round((chance94 + (float(tm['perc_43'][i]) +
                                   float(tm['perc_33'][i]) )/100),3)
    chance92 = round((chance93 + (float(tm['perc_32'][i]) +
                                   float(tm['perc_22'][i]) )/100),3)
    chance91 = round((chance92 + (float(tm['perc_21'][i]) +
                                   float(tm['perc_11'][i]) )/100),3)
    chances = [chance91, chance92, chance93,
               chance94, chance95, chance96,
```



```

        #calculates probabilites as a decimal from the sting
        stored in TM
        chances = chance_gcse(i)
        print('Chances: '+str(chances))
    else:
        #caclulates the output for combined-science which has a
        different data structure
        expected = round((float(tm['perc_11'][i])*1
                        + float(tm['perc_21'][i])*1.5
                        + float(tm['perc_22'][i])*2
                        + float(tm['perc_32'][i])*2.5
                        + float(tm['perc_33'][i])*3
                        + float(tm['perc_43'][i])*3.5
                        + float(tm['perc_44'][i])*4
                        + float(tm['perc_54'][i])*4.5
                        + float(tm['perc_55'][i])*5
                        + float(tm['perc_65'][i])*5.5
                        + float(tm['perc_66'][i])*6
                        + float(tm['perc_76'][i])*6.5
                        + float(tm['perc_77'][i])*7
                        + float(tm['perc_87'][i])*7.5
                        + float(tm['perc_88'][i])*8
                        + float(tm['perc_98'][i])*8.5
                        + float(tm['perc_99'][i])*9
                        )/100,2)

        #caluates cumulative probabilities of achieving a grade
        chances = chance_combined_science(i)
        print('Chances: '+str(chances))
    #displays probabilities of achieving threshold grades
    print('Chance 7+ ' + str(chances[6]*100) + '%')
    print('Chance 5+ ' + str(chances[4]*100) + '%')
    print('Chance 4+ ' + str(chances[3]*100) + '%')
    #statistics as a tuple
    stats =[expected, chances, band]

    return stats

#Access the latet DfE Transition Matix and store as a data-frame
SOURCE = "https://explore-education-statistics.service.gov.uk/" \
"data-catalogue/data-set/f65c2e07-a857-4245-b5ba-8effb0f89ba8/csv"
df = pd.read_csv(SOURCE)
#convert the data frame to a dictionary
tm=df.to_dict()

#Only executed if the script is run as a main programme
if __name__ == "__main__":
    #runs the code with test data
    SUBJECT='Art and Design'

```

```
#SUBJECT = 'Combined Science'  
KS2=102  
get_expected(SUBJECT,KS2)
```

Appendix 4 – Flake8 on Transition Matrix Code

(uni) PS D:\OneDrive - Smart MAT\2. Uni\Project Management> flake8 trans_matsv2.py

trans_matsv2.py:9:1: E302 expected 2 blank lines, found 1

trans_matsv2.py:13:26: E225 missing whitespace around operator

trans_matsv2.py:25:1: E302 expected 2 blank lines, found 1

trans_matsv2.py:25:28: E211 whitespace before '('

trans_matsv2.py:27:48: E231 missing whitespace after ','

trans_matsv2.py:29:33: E128 continuation line under-indented for visual indent

trans_matsv2.py:29:56: E202 whitespace before ')'

trans_matsv2.py:29:63: E231 missing whitespace after ','

trans_matsv2.py:31:33: E128 continuation line under-indented for visual indent

trans_matsv2.py:31:56: E202 whitespace before ')'

trans_matsv2.py:31:63: E231 missing whitespace after ','

trans_matsv2.py:33:33: E128 continuation line under-indented for visual indent

trans_matsv2.py:33:56: E202 whitespace before ')'

trans_matsv2.py:33:63: E231 missing whitespace after ','

trans_matsv2.py:35:33: E128 continuation line under-indented for visual indent

trans_matsv2.py:35:56: E202 whitespace before ')'

trans_matsv2.py:35:63: E231 missing whitespace after ','

trans_matsv2.py:37:33: E128 continuation line under-indented for visual indent

trans_matsv2.py:37:56: E202 whitespace before ')'

trans_matsv2.py:37:63: E231 missing whitespace after ','

trans_matsv2.py:39:33: E128 continuation line under-indented for visual indent

trans_matsv2.py:39:56: E202 whitespace before ')'

trans_matsv2.py:39:63: E231 missing whitespace after ','

trans_matsv2.py:41:33: E128 continuation line under-indented for visual indent

trans_matsv2.py:41:56: E202 whitespace before ')'

trans_matsv2.py:41:63: E231 missing whitespace after ','

trans_matsv2.py:43:33: E128 continuation line under-indented for visual indent

trans_matsv2.py:43:56: E202 whitespace before ')'

trans_matsv2.py:43:63: E231 missing whitespace after ','

trans_matsv2.py:49:1: E302 expected 2 blank lines, found 1

trans_matsv2.py:50:80: E501 line too long (95 > 79 characters)

trans_matsv2.py:51:10: E225 missing whitespace around operator

trans_matsv2.py:53:9: E265 block comment should start with '# '

trans_matsv2.py:58:5: E125 continuation line with same indent as next logical line

trans_matsv2.py:59:13: E265 block comment should start with '# '

trans_matsv2.py:61:13: E265 block comment should start with '# '

trans_matsv2.py:64:24: E225 missing whitespace around operator

trans_matsv2.py:66:17: E265 block comment should start with '# '

trans_matsv2.py:69:17: E265 block comment should start with '# '

trans_matsv2.py:73:13: E265 block comment should start with '# '

trans_matsv2.py:74:26: E225 missing whitespace around operator

trans_matsv2.py:74:51: E225 missing whitespace around operator

trans_matsv2.py:75:43: E225 missing whitespace around operator

trans_matsv2.py:75:63: E225 missing whitespace around operator

trans_matsv2.py:75:66: E225 missing whitespace around operator

trans_matsv2.py:75:80: E501 line too long (99 > 79 characters)

trans_matsv2.py:77:17: E265 block comment should start with '# '

trans_matsv2.py:79:21: E265 block comment should start with '# '

trans_matsv2.py:80:80: E501 line too long (89 > 79 characters)

trans_matsv2.py:81:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:81:80: E501 line too long (89 > 79 characters)

trans_matsv2.py:82:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:82:80: E501 line too long (89 > 79 characters)

trans_matsv2.py:83:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:83:80: E501 line too long (89 > 79 characters)

trans_matsv2.py:84:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:84:68: E231 missing whitespace after ','

trans_matsv2.py:85:21: E265 block comment should start with '# '

trans_matsv2.py:86:26: E211 whitespace before '('

trans_matsv2.py:87:21: E265 block comment should start with '# '

trans_matsv2.py:87:80: E501 line too long (85 > 79 characters)

trans_matsv2.py:91:21: E265 block comment should start with '# '

trans_matsv2.py:91:80: E501 line too long (100 > 79 characters)

trans_matsv2.py:93:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:94:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:95:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:96:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:97:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:98:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:99:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:100:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:101:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:102:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:103:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:104:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:105:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:106:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:107:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:108:37: E128 continuation line under-indented for visual indent

trans_matsv2.py:109:37: E124 closing bracket does not match visual indentation

trans_matsv2.py:109:42: E231 missing whitespace after '
trans_matsv2.py:111:21: E265 block comment should start with '# '
trans_matsv2.py:114:17: E265 block comment should start with '# '
trans_matsv2.py:118:17: E265 block comment should start with '# '
trans_matsv2.py:119:24: E225 missing whitespace around operator
trans_matsv2.py:122:1: E265 block comment should start with '# '
trans_matsv2.py:123:1: E305 expected 2 blank lines after class or function definition,
found 0
trans_matsv2.py:124:1: E122 continuation line missing indentation or outdented
trans_matsv2.py:126:1: E265 block comment should start with '# '
trans_matsv2.py:127:3: E225 missing whitespace around operator
trans_matsv2.py:129:1: E265 block comment should start with '# '
trans_matsv2.py:131:5: E265 block comment should start with '# '
trans_matsv2.py:132:12: E225 missing whitespace around operator
trans_matsv2.py:133:5: E265 block comment should start with '# '
trans_matsv2.py:134:8: E225 missing whitespace around operator
trans_matsv2.py:135:25: E231 missing whitespace after '

[illegible]

[illegible]

```

        + float(tm['perc_9'][i])*9
        )/100, 2
    )

    # display expected value
    print('Expected: ' + str(expected))
    # calculates probabilities as a decimal
    # from the sting stored in TM
    chances = chance_gcse(i)
    print('Chances: '+str(chances))
else:
    # caclulates the output for combined-science
    # which has a different data structure
    expected = round((
        float(tm['perc_11'][i])*1
        + float(tm['perc_21'][i])*1.5
        + float(tm['perc_22'][i])*2
        + float(tm['perc_32'][i])*2.5
        + float(tm['perc_33'][i])*3
        + float(tm['perc_43'][i])*3.5
        + float(tm['perc_44'][i])*4
        + float(tm['perc_54'][i])*4.5
        + float(tm['perc_55'][i])*5
        + float(tm['perc_65'][i])*5.5
        + float(tm['perc_66'][i])*6
        + float(tm['perc_76'][i])*6.5
        + float(tm['perc_77'][i])*7
        + float(tm['perc_87'][i])*7.5
        + float(tm['perc_88'][i])*8
        + float(tm['perc_98'][i])*8.5
        + float(tm['perc_99'][i])*9
        )/100, 2
    )

    # caluates cumulative probabilities of achieving a grade
    chances = chance_combined_science(i)
    print('Chances: '+str(chances))
    # displays probabilities of achieving threshold grades
    print('Chance 7+ ' + str(chances[6]*100) + '%')
    print('Chance 5+ ' + str(chances[4]*100) + '%')
    print('Chance 4+ ' + str(chances[3]*100) + '%')
    # statistics as a tuple
    stats = [expected, chances, band]

return stats

# Access the latet DfE Transition Matix and store as a data-frame
SOURCE = "https://explore-education-statistics.service.gov.uk/" \
"data-catalogue/data-set/f65c2e07-a857-4245-b5ba-8effb0f89ba8/csv"

```

```
df = pd.read_csv(SOURCE)
# convert the data frame to a dictionary
tm = df.to_dict()

# Only executed if the script is run as a main programme
if __name__ == "__main__":
    # runs the code with test data
    SUBJECT = 'Art and Design'
    # SUBJECT = 'Combined Science'
    KS2 = 102
    get_expected(SUBJECT, KS2)
```